Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 12: Reproduction in Plants

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	12 — Reproduction in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps target the most-confused pairs: pollination (pollen transfer) vs fertilisation (gamete fusion); ovule→seed vs ovary→fruit vs zygote→embryo; self- vs cross-pollination; bisexual vs unisexual; vegetative propagation (clone, one parent) vs seed reproduction (variation, two parents); spore (asexual unit, no gamete fusion) vs seed (post-fertilisation); budding vs fragmentation vs spore formation; matching each dispersal/pollination agent to the correct adaptation. Do not write on this paper — mark answers on your answer sheet.

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1. A nursery worker scrapes off every bud from a section of stored potato before planting it, leaving the rest of the tuber intact. Compared with an unscraped tuber, the scraped section will most likely:

(A) sprout even faster, because removing the buds frees stored food for the rest
(B) produce seeds instead of shoots, switching to sexual reproduction
(C) fail to sprout new shoots, because the buds (eyes) are what grow into new plants
(D) grow roots from the cut surface but still send up shoots from the food store

2. A gardener bends a low rose branch to the ground, covers a part of it with soil while it is still attached to the parent, and weeks later cuts off a rooted new plant. The technique used is:

(A) cross-pollination between the branch and the soil
(B) spore formation triggered by burying the stem
(C) fragmentation, because the branch was finally cut
(D) layering, a form of vegetative propagation

3. In grafting, a cut shoot of a desired variety is joined onto the rooted stem of another plant. The chief reason a gardener prefers grafting over growing the variety from seed is that grafting:

(A) keeps the new shoot genetically identical to the chosen parent variety
(B) guarantees the offspring will be a brand-new mix of two varieties' traits
(C) lets the plant skip pollination by making its own pollen sooner
(D) produces spores that survive drought better than seeds

4. A teacher places a fresh Bryophyllum leaf flat on moist soil. Tiny plantlets appear along its edge even though the leaf was never planted upright. This works because the leaf margin carries:

(A) seeds that were already inside the leaf from an earlier flower
(B) spores released onto the leaf by nearby fungi
(C) ovules that germinate once they touch moist soil
(D) buds that can each develop into a complete new plant

5. Onion and ginger are both used in the kitchen, yet a gardener can grow new plants from either by burying the underground part. The mode of reproduction this relies on is:

(A) spore formation, because the parts are kept underground in the dark
(B) sexual reproduction, since the buried part contains tiny ovules
(C) fragmentation, because the part is broken off before planting
(D) vegetative propagation from an underground stem/bud structure

6. Yeast in warm sugary dough multiplies very fast. Under the microscope a small outgrowth is seen bulging from a cell, narrowing at its base, and finally pinching off. The key feature that makes this budding rather than fragmentation is that:

(A) the parent cell breaks into several equal pieces that each regrow
(B) two yeast cells fuse before the new cell appears
(C) spores are released from a case on the parent
(D) a small bud grows out of one parent cell and separates as a new individual

7. A pond sample of the green alga Spirogyra is left in good light; within days each original filament has become several shorter filaments. The most likely explanation is that the alga reproduced by:

(A) budding, with each filament throwing out small buds
(B) spore formation inside a thick-walled case
(C) fragmentation, each broken piece growing into a new filament
(D) fertilisation between two filaments forming new ones

8. A mushroom, a moss and a fern are grouped together in a biology lesson because they share one reproductive feature. That shared feature is that they all:

(A) produce spores that are protected in spore cases until conditions are right
(B) grow new individuals from buds on their leaf margins

(C) form seeds inside fruits after pollination(D) reproduce only by stem cuttings

9. Spores can wait through a hot dry spell and then sprout after the first rains. The chief survival value of this property is that spores:

(A) are heavy, so they drop straight down and never dry out

(B) must combine with another spore before they can survive a drought

(C) germinate immediately whatever the weather, before they can dry

(D) stay protected in their cases through bad conditions and germinate only when it turns moist

10. A student lists four plants and how each is usually grown: mango from seed, sugarcane from stem cutting, money plant from stem cutting, rose from stem cutting. Which single plant in the list is the odd one out by mode of reproduction?

(A) sugarcane, because cuttings cannot really root

(B) mango, because it is grown from a seed (sexual), the rest by vegetative propagation

(C) money plant, because it grows only in water and not by cuttings

(D) rose, because roses reproduce only by spores

11. Two beds of the same crop are planted: Bed A from cuttings of one parent, Bed B from seeds of that parent. After a dry, disease-prone season, a key advantage Bed B may show over Bed A is that Bed B:

(A) is made of exact clones, giving it more uniform resistance than Bed A

(B) grew without any parent at all, so it carries no inherited weakness

(C) reproduced by spores, which always survive drought better than cuttings

(D) contains varied offspring, so some plants may better withstand the new conditions

12. Why does growing a fruit variety only from cuttings, generation after generation, give no new improved varieties, while growing it from seeds occasionally does?

(A) cuttings make clones with no new mixing of traits, whereas seeds come from fertilisation that can produce variation

(B) cuttings always produce spores that lose the parent's good traits

(C) seeds skip fertilisation, so they invent traits the parent never had

(D) cuttings need two parents while seeds need only one

13. A child argues that 'all reproduction in plants must involve a flower'. The best single counter-example showing this is wrong is:

(A) a mango tree, which flowers before it fruits

(B) an insect-pollinated rose, whose bright flower attracts bees

(C) a fern, which makes new ferns from spores and never produces a flower

(D) a papaya, which bears separate male and female flowers

14. A gardener wants a quick, reliable supply of new money plants over winter. Which method best suits the plant's natural way of multiplying?

(A) collect its seeds and sow them in trays

(B) cut stem pieces and let each root into a new plant

(C) wait for insects to cross-pollinate the flowers

(D) scatter its spores onto moist soil

15. Examined under a hand lens, a single stamen of a flower shows a slender stalk topped by a knob-like sac. The knob-like sac and the slender stalk are, respectively, the:

(A) stigma and the style
(B) ovary and the ovule
(C) anther and the filament
(D) filament and the anther

16. Tracing the female parts of a flower from the receptive tip down to where future seeds form, the correct order is:

(A) anther, then filament, then ovary
(B) stigma, then style, then ovary
(C) ovary, then style, then stigma
(D) stigma, then ovary, then style

17. A flower is found to make pollen but has no stigma, style or ovary of its own. This flower is best described as:

(A) a bisexual flower
(B) a unisexual male flower
(C) a unisexual female flower
(D) a spore-bearing flower

18. If the entire ovary were carefully removed from a flower before pollination, which later event becomes impossible?

(A) formation of seeds, because the ovules that turn into seeds were inside the ovary

(B) production of pollen, because pollen is made in the ovary

(C) the stigma receiving pollen, because the stigma sits on the ovary

(D) wilting of the petals, because petals grow from the ovary

19. A flower has bright petals, a pistil, and stamens all in the same bloom. Without watching any animal, what can you already conclude about its potential for pollination?

(A) only cross-pollination is possible because one flower cannot pollinate itself

(B) no pollination can occur until it grows a second flower

(C) it must reproduce by spores since it has both parts

(D) self-pollination is possible because pollen and stigma are present together in one flower

20. In a maize cob the grains form on a structure quite separate from the tassel that sheds pollen at the top of the plant. The most reasonable interpretation is that maize has:

(A) a single bisexual flower carrying every grain

(B) separate male and female (unisexual) flowers on the same plant

(C) no flowers at all, reproducing only by spores

(D) flowers that make seeds without any pollen

21. Why is the anther, rather than the stigma, the part that must be dusted to collect pollen for a school experiment?

(A) the stigma makes the pollen and the anther only stores ovules

(B) the anther is where pollen is actually produced, while the stigma only receives it

(C) both make pollen, but the stigma's is too sticky to collect

(D) neither makes pollen; pollen comes from the ovary

22. A diagram labels four parts of one flower: A makes pollen, B is the sticky tip, C is the stalk holding A, D contains the ovules. The correct names for A, B, C, D are:

(A) A = stigma, B = anther, C = style, D = ovary

(B) A = anther, B = stigma, C = filament, D = ovary

(C) A = anther, B = filament, C = stigma, D = ovary

(D) A = ovary, B = stigma, C = filament, D = anther

23. A flower has stamens but its pistil is sterile and bears no ovules at all. Even with perfect pollination, this flower cannot:

(A) release pollen, because the missing ovules block the anther

(B) have its stigma receive pollen, because no ovules means no stigma

(C) form seeds, because there are no ovules to become seeds after fertilisation

(D) be called unisexual, because it still has a pistil shape

24. A student writes: 'The filament is the part on which pollen lands.' Why is this statement incorrect?

(A) pollen actually lands on the anther, not the filament

(B) pollen lands inside the ovary, not on the filament

(C) the statement is correct, because the filament is sticky and catches pollen

(D) pollen lands on the stigma; the filament is only the stalk that holds the anther aloft

25. Pollen from the anther of one mustard flower is carried to the stigma of a different mustard plant. This event, in itself, is correctly called:

(A) fertilisation

(B) self-pollination

(C) cross-pollination

(D) seed dispersal

26. Two events are described: (i) pollen lands on a stigma; (ii) a male gamete fuses with the egg in the ovule. Which statement about their relationship is correct?

(A) (i) is pollination and must happen before (ii), the fertilisation

(B) (i) and (ii) are the same event under two names

(C) (ii) happens first and triggers (i)

(D) (i) is fertilisation and (ii) is pollination

27. After pollen settles on the stigma, by what route does the male gamete reach the egg in the ovule?

(A) a pollen tube grows down through the style and into the ovary to the ovule

(B) the gamete drips down the outside of the filament

(C) the gamete is blown by wind straight into the ovary

(D) the stigma swallows the pollen and pushes it through a petal

28. A flower is visited by bees and its stigma is well dusted with pollen, yet weeks later no fruit develops. The most reasonable single explanation is that:

(A) pollination clearly failed, since pollen never reached the stigma

(B) the ovary turned directly into seeds without becoming a fruit

(C) fertilisation did not occur — the male gamete never fused with the egg in the ovule

(D) seed dispersal happened before the fruit could form

29. Immediately after the male gamete fuses with the egg, the very first structure formed is the:

(A) fruit (B) seed coat

(C) zygote (D) pollen grain

30. After fertilisation in a flowering plant, match each starting part to what it becomes: ovule, ovary, zygote.

(A) ovule → fruit, ovary → seed, zygote → embryo

(B) ovule → seed, ovary → fruit, zygote → embryo

(C) ovule → seed, ovary → embryo, zygote → fruit

(D) ovule → embryo, ovary → fruit, zygote → seed

31. A botanist cuts open a single fruit and counts seven seeds inside it. The most direct conclusion about the original flower is that its ovary held:

(A) seven anthers, one for every seed

(B) at least seven ovules, each of which became a seed after fertilisation

(C) exactly one ovule that split into seven seeds

(D) seven stigmas that each grew into a seed

32. Soon after a pea flower is fertilised, the petals and stamens shrivel and drop, while the base of the flower begins to swell. The swelling part is the:

(A) anther, which is developing into the seeds(B) stigma, which is swelling into the fruit

(C) filament, which becomes the pod

(D) ovary, which is developing into the fruit (the pod)

33. Arrange these stages of a flowering plant's life in the correct order: (i) zygote forms; (ii) pollen reaches the stigma; (iii) seed grows into a new plant; (iv) ovary ripens into a fruit with seeds.

(A) (i) → (ii) → (iii) → (iv) (B) (ii) → (i) → (iv) → (iii)

(C) (ii) → (iv) → (i) → (iii) (D) (iv) → (ii) → (i) → (iii)

34. Apple flowers depend on bees, while the flowers of many grasses are small, dull-green and scentless. The simplest reason grass flowers lack bright petals and scent is that they:

(A) are pollinated by wind, which is not attracted by colour or smell

(B) are pollinated by insects that prefer dull, scentless flowers

(C) do not need pollination because they reproduce by spores
(D) are pollinated by water flowing over the soil

35. Two flowers are compared: Flower X is large, brightly coloured, scented and produces sticky pollen; Flower Y is small, drab, scentless and produces huge amounts of light, dry pollen. Their likely pollinating agents are:

(A) Flower X by insects and Flower Y by wind
(B) Flower X by wind and Flower Y by insects(C) both flowers by water
(D) both flowers by insects only

36. Why does a typical wind-pollinated flower produce far more pollen than an insect-pollinated flower of similar size?

(A) wind carries pollen directly to the right stigma, so extra is wasted on purpose
(B) wind pollen is heavier, so more is needed to make it fly
(C) insects eat most of the wind-flower's pollen before it lands
(D) wind scatters pollen randomly, so a huge surplus is needed for any to chance upon a stigma

37. An aquatic plant releases pollen that is carried along by the flow of a stream to reach other flowers. Its pollinating agent is:

(A) wind (B) water
(C) insects attracted by nectar (D) animals that eat the flowers

38. A flower offers a sweet scent and a pool of nectar deep inside, with landing-pad petals. These features are best explained as adaptations to attract:

(A) insects, which visit for nectar and carry pollen between flowers
(B) wind, which is drawn in by the scent
(C) water, which dissolves the nectar and spreads pollen
(D) fungi, which feed on the nectar

39. A grower wants to be sure a particular tomato flower is pollinated by its own pollen and not by a neighbour's. She covers the unopened flower with a fine bag so no insect or breeze can enter, yet it still sets fruit. This shows the flower can undergo:

(A) cross-pollination, since the bag let neighbour pollen slip in
(B) fertilisation without any pollination at all
(C) self-pollination, using its own pollen on its own stigma
(D) spore formation triggered by the dark bag

40. Which statement correctly distinguishes self-pollination from cross-pollination?

(A) self-pollination always needs an insect, while cross-pollination needs none
(B) self-pollination keeps pollen within the same flower or plant; cross-pollination carries it to a different plant of the same kind

(C) self-pollination produces spores, cross-pollination produces seeds
(D) cross-pollination keeps pollen on the same flower; self-pollination sends it away

41. A gardener notices a flower that opens, sheds pollen onto its own stigma before any insect arrives, and sets seed. The chief advantage of this self-pollination for the plant is that it:

(A) can reproduce reliably even when no pollinating agent or partner is available
(B) guarantees lots of variation among its offspring
(C) lets two different parents combine their best traits
(D) avoids the need to form a zygote

42. In an experiment, the anthers are snipped off a young bisexual flower before it opens, and it is then left uncovered near other plants of the same kind. If it later sets seed, the pollination responsible must have been:

(A) self-pollination, using the flower's own remaining pollen
(B) no pollination, since seeds can form without pollen
(C) cross-pollination, because its own pollen source was removed
(D) spore formation by the anther-less flower

43. A poster claims, 'Pollination by an insect is the moment the new plant begins to form.' What is the precise correction?

(A) the new plant begins the instant pollen leaves the anther
(B) the poster is right; pollination and the start of the new plant are the same event
(C) the new plant begins only when the fruit is eaten and the seed dispersed
(D) the new plant begins at fertilisation, when gametes fuse to form the zygote, which happens after pollination

44. A dandelion seed carries a tuft of fine hairs like a tiny parachute. This structure is an adaptation for dispersal by:

(A) water, which the hairs help it float on(B) animals, which the hairs help it stick to
(C) explosion, which the hairs help trigger
(D) wind, which catches the hairs and carries the light seed far

45. A fruit has a tough rind enclosing a thick, air-filled spongy layer and is found bobbing along a river. These features point to dispersal by:

(A) wind, because the rind makes it light enough to fly
(B) animals, because the spongy layer is tasty to eat
(C) water, because the spongy buoyant layer lets it float and be carried by currents
(D) explosion, because the rind bursts when ripe

46. Walking through tall grass, a hiker finds many small fruits with stiff hooks clinging to her socks. This shows the plant is adapted for dispersal by:

(A) wind, because hooks help the fruit fly(B) water, because hooks help the fruit float

(C) animals, because the hooks catch onto fur or clothing and are carried away

(D) explosion, because hooks make the fruit burst

47. Why are juicy, brightly coloured, sweet fruits considered an adaptation for dispersal by animals?

(A) the bright colour makes the fruit light enough to be blown by wind

(B) the juice helps the whole fruit float away on rivers

(C) the sweetness makes the fruit burst and fling its seeds

(D) animals eat the fruit and later drop or pass the tough seeds far from the parent

48. A pod dries in the sun until it suddenly splits with a snap and throws its seeds a short distance around the plant. This method of dispersal is by:

(A) wind catching the seeds as they leave(B) animals carrying the seeds in the pod

(C) explosion (sudden bursting of the dried fruit)

(D) water washing the seeds out of the pod

49. A plant always drops its seeds straight to the ground at its own base, with no special feature for travelling. A likely long-term disadvantage of this is that:

(A) the crowded seedlings compete with the parent and each other for light, water, minerals and space

(B) the seedlings cannot photosynthesise so close to the parent

(C) the seeds will turn back into spores if they land near the parent

(D) the seedlings get far too much space and starve from being too spread out

50. Beyond easing overcrowding, what further benefit does seed dispersal give a plant species?

(A) it lets the plant colonise new areas where conditions may be favourable

(B) it makes each seed heavier so it stays put

(C) it speeds up the pollination of the parent flower

(D) it removes the need for fertilisation in the next generation

51. Hydra, a tiny water animal, grows a bulge that enlarges and detaches as a new individual. Which plant-kingdom organism reproduces by the same basic process?

(A) yeast, which buds off new cells (B) Spirogyra, which breaks into fragments

(C) a fern, which scatters spores

(D) a potato, which sprouts from buds in its eyes

52. Which pairing of an organism with its mode of reproduction is correct?

(A) yeast — fragmentation (B) Spirogyra — budding

(C) moss — spore formation (D) Bryophyllum — spore formation

53. Which single observation would most convincingly prove that a new plant arose by sexual reproduction rather than vegetative propagation?

(A) it grew from a seed that had formed inside a fruit after a flower was fertilised

(B) it grew from a buried piece of underground stem

(C) it sprouted from a bud on a leaf margin

(D) it rooted from a stem cutting placed in water

54. A spore and a seed are sometimes confused. The most fundamental difference is that:

(A) a spore forms after fertilisation, whereas a seed needs no gametes

(B) a seed forms after fertilisation (fusion of gametes), whereas a spore is an asexual unit formed without gamete fusion

(C) both form only after pollination by insects

(D) a seed is always lighter and more wind-blown than a spore

55. A banana variety grown by farmers makes no usable seeds, yet new banana plants keep appearing. The best explanation is that the banana is propagated:

(A) by spores released from the ripe fruit(B) by wind-blown seeds too small to notice

(C) vegetatively, from its underground stem, so no seeds are needed

(D) by cross-pollination that hides the seeds inside

56. A field of a crop grown entirely from cuttings of a single parent is struck by a new disease, and almost every plant falls ill at once. The most likely reason the disease spread so uniformly is that:

(A) the plants vary widely, so the disease found many weak points

(B) the plants reproduced sexually, mixing in the disease

(C) the plants formed spores that carried the disease

(D) the plants are clones with identical traits, so they share the same vulnerability

57. A flower is successfully cross-pollinated, fertilisation occurs, and a fruit forms. Tracing one specific ovule, its correct fate is that it becomes a:

(A) fruit that encloses the other ovules

(B) seed, inside which the zygote it held has grown into an embryo

(C) new stigma for the next flower

(D) pollen grain for the next round of pollination

58. A teacher says a fern growing on a damp wall and a potato sprouting in a dark cupboard reproduce in 'completely different ways'. The sharpest way to state that difference is:

(A) the fern reproduces sexually by seeds, the potato by spores

(B) both reproduce by budding, just in different places

(C) the fern reproduces by stem cuttings, the potato by fragmentation

(D) the fern reproduces by spores (asexual, via spore cases), while the potato reproduces vegetatively from buds in its tuber

59. Why must pollination, not fertilisation, be the step an insect performs when it carries pollen between two flowers?

(A) the insect only moves pollen to the stigma; the actual gamete fusion happens later inside the ovule

(B) the insect fuses the gametes itself while feeding on nectar

(C) the insect carries finished seeds, not pollen, between flowers

(D) fertilisation is just another word for an insect's visit

60. A student classifies four examples of new-plant formation: (1) Bryophyllum leaf-margin buds, (2) ginger underground stem, (3) yeast bud cells, (4) a mango seed. Which one does NOT belong with the other three?

(A) the yeast bud, because budding is the only sexual process listed

(B) the mango seed, because it forms by sexual reproduction while the other three are asexual

(C) the ginger stem, because underground parts cannot reproduce

(D) the Bryophyllum buds, because leaves never produce new plants

Solutions — Science (Class 7), Chapter 12: Reproduction in Plants — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	B	31	B	41	A	51	A
2	D	12	A	22	B	32	D	42	C	52	C
3	A	13	C	23	C	33	B	43	D	53	A
4	D	14	B	24	D	34	A	44	D	54	B
5	D	15	C	25	C	35	A	45	C	55	C
6	D	16	B	26	A	36	D	46	C	56	D
7	C	17	B	27	A	37	B	47	D	57	B
8	A	18	A	28	C	38	A	48	C	58	D
9	D	19	D	29	C	39	C	49	A	59	A
10	B	20	B	30	B	40	B	50	A	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) In vegetative propagation by tubers, the new shoots arise from the buds sitting in the 'eyes' of the potato. Remove the buds and there is nothing left to grow a shoot, so the scraped section stays dormant. It cannot sprout faster without buds; tubers do not switch to making seeds; and the starchy store is food, not a growing point, so shoots will not appear from it alone.

2. (D) Encouraging a still-attached stem to root while pressed into soil, then severing it, is layering — a vegetative (asexual) method giving a clone of the parent. No pollen or gametes are involved, so it is not pollination. Spores are not formed by burying a stem. Fragmentation refers to an organism breaking into pieces that each regrow (as in Spirogyra), not a deliberate bend-and-bury of one branch.

3. (A) Grafting is vegetative propagation, so the joined shoot grows as a clone of the chosen variety and keeps its exact qualities. Seeds, by contrast, come from fertilisation and can vary. Grafting does not create a new trait mix (that would be sexual), it is unrelated to making pollen, and it produces no spores.

4. (D) Bryophyllum reproduces vegetatively from buds carried in the notches of its leaf margins; in moist conditions each bud grows into a new plant. These are buds, not seeds (no fertilisation happened), not fungal spores, and not ovules (ovules sit inside an ovary and become seeds only after fertilisation).

5. (D) Onion (a bud-bearing structure) and ginger (an underground stem) both sprout new shoots from their buds — vegetative propagation, an asexual method giving clones. Being underground does not make them form spores; they contain buds, not ovules, so no sexual reproduction occurs; and simply detaching a part for planting is not the regrow-from-pieces process of fragmentation.

6. (D) In budding, a single small outgrowth (bud) forms on the parent, enlarges, and detaches — exactly what is described. Fragmentation is the parent breaking into several pieces, which is not happening here. No fusion of cells occurs (that would be sexual), and no spore case is involved.

7. (C) Spirogyra commonly reproduces by fragmentation: the filament breaks into pieces and each fragment grows into a full new filament, which is why one becomes many. It does not bud off small outgrowths the way yeast does, the multiplication here needs no spore case, and no gamete fusion is described.

8. (A) Fungi (like mushrooms), mosses and ferns all reproduce by spore formation; the spores are kept safe in spore cases and germinate when it is moist. They do not bear leaf-margin buds (that is Bryophyllum), they do not make seeds or fruits (those need flowers and fertilisation), and they are not propagated by cuttings.

9. (D) A spore's tough case lets it survive unfavourable conditions and germinate only when moisture returns — a built-in timing advantage. Spores are actually light and spread by wind, not heavy droppers; a single spore can grow on its own without combining with another; and germinating in any weather would expose the new growth to drought, which is the opposite of an advantage.

10. (B) Sugarcane, money plant and rose are all routinely grown from stem cuttings — vegetative (asexual) propagation. Mango is grown from a seed, which forms after fertilisation, so it is the sexual-reproduction odd one out. Sugarcane cuttings do root; money plant grows readily from stem cuttings (water just helps rooting); and roses are not spore-formers.

11. (D) Bed B came from seeds, the product of sexual reproduction with two parents' gametes, so its plants vary — and variation means some may survive a new stress that wipes out identical plants. Bed A's cuttings are clones, so they are uniform, not more varied; both beds do have parents; and neither bed was grown from spores.

12. (A) Cuttings are vegetative propagation: every offspring is a clone of one parent, so no new trait combinations appear. Seeds form after fertilisation, which mixes gametes from parents and can throw up variation — the raw material for new varieties. Cuttings do not make spores, seeds do not skip fertilisation, and it is cuttings (not seeds) that need only one parent.

13. (C) A fern reproduces by spores and bears no flowers at all, directly disproving the claim. The mango, rose and papaya examples all involve flowers, so they support rather than refute the child — they cannot serve as counter-examples.

14. (B) Money plant multiplies readily by stem cuttings (vegetative propagation), each piece rooting into a clone — quick and reliable. It is not usually grown from seed, it does not depend on insect pollination for propagation here, and it is a flowering plant, not a spore-former.

15. (C) A stamen is the anther (the sac that holds pollen) sitting on a filament (the slender stalk). The knob-like sac is therefore the anther and the stalk is the filament. Stigma and style belong to the pistil; ovary and ovule are female parts; and the last option reverses the two parts.

16. (B) The pistil runs stigma (the sticky tip that receives pollen) → style (the connecting stalk) → ovary (the base, which holds the ovules and later becomes the fruit). Anther and filament are male parts, not female; the second wrong option reverses top and bottom; and the last wrongly puts the ovary above the style.

17. (B) Having stamens (which make pollen) but lacking the pistil parts (stigma, style, ovary) makes it a male flower, and a flower with only one sex is unisexual. A bisexual flower would have both stamens and a pistil; a female flower would lack the pollen-making stamen; and flowers do not bear spores.

18. (A) The ovary holds the ovules, and ovules become seeds after fertilisation — remove the ovary and no seeds can form. Pollen is made in the anther, not the ovary; the stigma sits at the top of the pistil and can still receive pollen; and petals are separate floral parts that do not grow out of the ovary.

19. (D) When a single bisexual flower carries both the pollen-bearing anthers and a stigma, its own pollen can reach its own stigma — so self-pollination is possible. Such a flower is not limited to cross-pollination, it does not need a second flower for pollination to be possible, and possessing both sexes does not make a flower a spore-former.

20. (B) The pollen-shedding tassel and the grain-bearing cob are separate male and female flowering structures, so maize bears unisexual flowers. It is not a single bisexual flower; maize is a flowering plant, not a spore-former; and seeds (grains) form only after pollen and fertilisation, never without pollen.

21. (B) Pollen is produced in the anther; the stigma is merely the landing surface that receives pollen during pollination. So you collect pollen from the anther. The stigma does not make pollen, the anther does not store ovules, and pollen is never produced in the ovary (that is where ovules sit).

22. (B) Pollen is made in the anther (A), the sticky receiving tip is the stigma (B), the stalk carrying the anther is the filament (C), and the part holding ovules is the ovary (D). The other sets misplace at least one part — e.g. swapping anther with stigma, or calling the pollen-maker the ovary.

23. (C) Seeds develop from ovules after fertilisation; with no ovules, no seeds can ever form even if pollination succeeds. The missing ovules do not stop the anther from shedding pollen, nor do they erase the stigma (which can still receive pollen). The naming point is irrelevant to whether seeds can form.

24. (D) During pollination, pollen lands on the stigma, the receptive tip of the pistil. The filament's job is simply to hold the anther up, not to catch pollen. Pollen also does not land on the anther (the anther produces it) nor inside the ovary, and the filament is not a sticky catching surface.

25. (C) Pollen moving to the stigma of a separate plant of the same kind is cross-pollination. It is not fertilisation, which is the later fusion of gametes inside the ovule. It is not self-pollination (that stays within the same flower or plant), and it is certainly not seed dispersal, which moves finished seeds.

26. (A) Landing of pollen on the stigma is pollination (i); fusion of the male gamete with the egg is fertilisation (ii). Pollination comes first, because the pollen must arrive before its gamete can reach the egg. The two are different events, fertilisation cannot precede pollination, and the names are not swapped.

27. (A) The pollen grain grows a pollen tube that travels down the style into the ovary, delivering the male gamete to the egg in the ovule. The gamete does not run down the filament (a male stalk on the other side of the flower), it is not wind-blown into a sealed ovary, and there is no passage through a petal.

28. (C) Pollination plainly succeeded (the stigma was dusted), but a fruit forms only after fertilisation; if no gamete fusion followed, no fruit develops. The first option contradicts the given fact that pollen reached the stigma. The ovary becomes the fruit (not directly the seeds), and dispersal moves seeds that exist only after a fruit has formed.

29. (C) The fusion of the male gamete with the egg produces a zygote, the first cell of the new individual, which later grows into the embryo. The fruit develops later from the ovary, the seed coat forms as the ovule matures into a seed, and the pollen grain existed before fertilisation, not as a product of it.

30. (B) The standard outcomes are: each ovule becomes a seed, the whole ovary ripens into the fruit, and the zygote grows into the embryo inside the seed. The wrong options swap these — for example making the ovule into a fruit or the zygote into a seed — which contradicts the actual development.

31. (B) Each seed develops from a fertilised ovule, so seven seeds mean the ovary contained at least seven ovules. Anthers are male parts unrelated to seed number; a single ovule becomes one seed, not seven; and stigmas are receiving surfaces, not seed-forming structures.

32. (D) After fertilisation the ovary enlarges into the fruit (in peas, the pod), while the now-useless petals and stamens wither and fall. The anther is a male part that does not become seeds, the stigma is a thin receptive tip that does not swell into a fruit, and the filament is just a stalk.

33. (B) Pollination (ii) comes first; it allows fertilisation, which forms the zygote (i); then the ovary ripens into a fruit holding seeds (iv); finally a dispersed seed germinates into a new plant (iii). The other orders put the zygote before pollination, or the fruit before the zygote, both of which are impossible.

34. (A) Bright colour, scent and nectar are advertisements for animal pollinators; wind cannot be tempted by them, so wind-pollinated grasses save on these features and instead make lots of light pollen. Insects are not drawn to dull, scentless flowers; grasses are flowering plants, not spore-formers; and they are not water-pollinated land plants.

35. (A) Bright colour, scent and sticky pollen are adaptations to attract and stick to insects (Flower X); small, drab, scentless flowers with abundant light, dry pollen suit dispersal by wind (Flower Y). The first wrong option reverses the agents; the others ignore the clear differences by assigning both to one agent.

36. (D) Wind delivers pollen blindly, so most grains never reach a suitable stigma; producing a vast surplus raises the odds that some succeed. Wind does not aim pollen accurately, wind-borne pollen is light (not heavy) so it can travel, and the explanation is not about insects eating the pollen of a wind-pollinated flower.

37. (B) When pollen is moved to the stigma by flowing or floating along water, the agent is water. Here the stream's flow carries the pollen, so it is water-pollinated, not wind-, insect- or animal-pollinated. The other agents are not involved in this water-borne transfer.

38. (A) Scent, nectar and a convenient landing surface are classic lures for insect pollinators, which pick up and transfer pollen while feeding. Wind is not attracted by scent or nectar, water does not respond to lures, and fungi are not pollinators of flowers.

39. (C) Bagging blocks outside pollen, so any fruit set must come from the flower's own pollen reaching its own stigma — self-pollination. The bag actually prevents cross-pollination, fertilisation still required pollination to occur first, and a bisexual flower does not switch to making spores in the dark.

40. (B) The difference is where the pollen ends up: within the same flower/plant (self) versus on a different plant of the same kind (cross). Neither type is defined by needing an insect, neither produces spores (both lead to seeds via fertilisation), and the last option reverses the two definitions.

41. (A) Self-pollination lets a plant set seed on its own, without waiting for wind, water, an insect or another plant — a reliable backup. It does not boost variation (that is a strength of cross-pollination with two parents), it involves only one parent, and it still leads to fertilisation and a zygote.

42. (C) Removing the anthers leaves the flower without its own pollen, so any pollen reaching its stigma had to come from another plant — cross-pollination. Self-pollination is ruled out because there is no own pollen; seeds cannot form without pollination; and an anther-less flower does not make spores.

43. (D) Pollination only delivers pollen to the stigma; the new individual actually begins as the zygote formed at fertilisation, a later step. So it is fertilisation, not pollination, that starts the new plant. Pollen leaving the anther is even earlier, the poster's equation is wrong, and dispersal happens long after the new plant (embryo) has already begun.

44. (D) A parachute of fine hairs increases air resistance so the light seed drifts on the breeze — a wind-dispersal adaptation. The hairs are not for floating on water, they are not hooks for clinging to animals, and they have nothing to do with a fruit bursting open.

45. (C) A buoyant, air-filled spongy layer keeps the fruit afloat so water currents can carry it — a water-dispersal adaptation. A tough rind does not make a fruit fly on wind; the spongy buoyancy layer is for floating, not as bait; and there is no mention of the fruit bursting.

46. (C) Hooks that latch onto fur or clothes let the fruit hitch a ride on a passing animal — animal dispersal. Hooks add weight and drag, hindering flight rather than helping it; they do not aid floating; and they are unrelated to a fruit bursting open.

47. (D) Tempting fruits get eaten by animals, which then carry the hardy seeds away in their gut or by dropping them, spreading the plant. Colour and juice do not lighten a fruit for wind, juiciness is not primarily a floatation aid, and sweetness does not cause bursting.

48. (C) When a drying fruit builds up tension and bursts open to fling its seeds, the dispersal is by explosion. Although the seeds become airborne briefly, the launching force is the bursting pod, not the wind; no animal carried them; and no water was involved.

49. (A) Without dispersal, seedlings pile up beside the parent and fight over the same light, water, minerals and space — overcrowding and competition. They can still photosynthesise; seeds do not revert to spores; and the problem is too little room, not too much, so 'too spread out' is the opposite of what happens.

50. (A) Carrying seeds away lets a species reach and settle new, possibly better, places — colonising new ground. Dispersal does not make seeds heavier (often the reverse), it happens long after the parent's pollination, and it does not abolish the need for fertilisation in future generations.

51. (A) Hydra and yeast both reproduce by budding — an outgrowth that grows and separates. Spirogyra uses fragmentation, a fern uses spore formation, and although a potato has 'buds', it propagates vegetatively from a tuber rather than by pinching off a bud-cell the way budding does.

52. (C) Moss reproduces by spore formation, so that pairing is correct. Yeast reproduces by budding (not fragmentation), Spirogyra by fragmentation (not budding), and Bryophyllum by buds on its leaf margins (vegetative propagation, not spores).

53. (A) A seed forms only after fertilisation, the hallmark of sexual reproduction, so growing from a seed is the convincing proof. Sprouting from an underground stem, a leaf-margin bud or a stem cutting are all vegetative (asexual) routes that need no fertilisation.

54. (B) A seed is the product of fertilisation — gametes fuse, an embryo forms inside — while a spore is an asexual reproductive unit produced without any gamete fusion. The first wrong option reverses this; spores are not made by insect pollination; and spores, not seeds, tend to be the very light wind-blown units.

55. (C) Seedless banana plants are multiplied vegetatively from the underground stem, an asexual method that needs no seeds at all. Bananas do not make spores from their fruit, they do not rely on tiny hidden seeds, and cross-pollination would still depend on seeds, which this variety lacks.

56. (D) Cuttings give clones — genetically identical plants — so if one is susceptible, all are, and a single disease can sweep through them uniformly. They do not vary widely (that is a feature of seed-grown crops), they were not grown sexually, and cuttings do not form spores.

57. (B) A fertilised ovule develops into a seed, and the zygote within it grows into the embryo. The fruit comes from the whole ovary, not from one ovule; an ovule does not turn into a stigma; and it certainly does not become pollen, which is a male product made in the anther.

58. (D) A fern makes spores held in spore cases, while a potato sprouts vegetatively from the buds in its eyes — two distinct asexual routes. The fern does not make seeds; neither uses budding in the yeast sense; and the fern is not grown from cuttings nor the potato by fragmentation.

59. (A) An insect's role is transport: it deposits pollen on the stigma, which is pollination. Fertilisation — the fusion of the male gamete with the egg — takes place later, deep inside the ovule, with no help from the insect. The insect does not fuse gametes, it carries pollen (not seeds), and an insect visit is not the same as fertilisation.

60. (B) Bryophyllum leaf buds, ginger's underground stem and yeast budding are all asexual (no gamete fusion); a mango seed forms only after fertilisation, so it is the sexual odd one out. Yeast budding is asexual, not sexual; ginger's underground stem clearly does reproduce vegetatively; and Bryophyllum proves that leaves can indeed give rise to new plants.